1. S3 Appendix. Open-data pipelines and reproducibility details

1.1. Scope

The open-data results summarized here are supplementary exploratory observable bridges, not part of the manuscript's primary simulator-first evidence ladder. They are intended as downstream signature tests of a latent structural-accessibility hypothesis. They do not directly observe M_b or any microscopic write variable. Their value lies instead in using fixed, inspectable pipelines to ask whether the predicted observable consequences survive in real neural recordings. Reviewers who only need to assess the main article claims can stop after the `E017--E022R` simulator stack described in the main text and reviewer notebook.

1.2. Public code and data record

All author-generated code, scripts, configuration files, and manuscript-facing outputs associated with these pipelines are publicly available in the `cytodendaccessmodel` GitHub repository (`https://github.com/NoeticDiffusion/cytodendaccessmodel`) and in the manuscript-matched Zenodo snapshot (DOI `10.5281/zenodo.19498499`; `https://doi.org/10.5281/zenodo.19498499`). The code is released under the GNU General Public License v3.0 (`GPL-3.0`). Third-party neural data are not redistributed in this repository; they remain accessible from their original DANDI Archive records (`000718`, `000336`, `001710`), while the derived manuscript-facing outputs discussed here are written under `data/dandi/triage/000718`, `data/dandi/triage/000336`, and `data/dandi/triage/001710`. Software identifiers, local build versions, and the consolidated computational reporting conventions shared with the simulator stack are summarized in S4 Appendix rather than duplicated here.

1.3. Null constructions and empirical controls

Because the open-data results are intended as downstream signature tests rather than direct observations of the latent variable M_b, the exact null architecture matters. The three dataset families therefore use explicit empirical nulls rather than a single generic asymptotic test.

For DANDI `000718`, the primary session-level enrichment summary compares each detected offline event against ten duration-matched inter-event windows drawn from the same offline session. At the event level, the Preferential Reactivation Index also uses a size-matched random-core null (`null_n = 500`) so that "more active than expected" means more active than a random set of equally many registered units, not merely more active than zero. Specificity is then stressed by registration-shuffle controls that permute the registered offline unit identities while leaving event structure intact (`20` repeats in the main PRI script).

For DANDI `000336`, the null is built within each analysis window by circularly shifting one imaging plane relative to the other (`n = 200` draws), preserving each plane's marginal activity and autocorrelation structure while breaking precise cross-plane alignment. Window-local null distributions are then aggregated with window-length weighting to produce the reported condition-level `z` versus null summaries.

For DANDI `001710`, the main inferential step is a subject-level permutation null (`n_perms = 1000`). Subject summaries are pooled, reassigned without replacement while preserving the observed group sizes, and compared by the difference in group means. The reported empirical p-value is one-sided in the observed direction, reflecting the planned claim boundary (`SparseKO < comparator`) rather than a two-sided search for any deviation whatsoever.

1.4. DANDI `000718`: Offline core-unit enrichment pipeline

The `000718` analysis [1] targets a narrow question: are the units most central to a Neutral-Exposure ensemble preferentially reactivated during later high-synchrony offline events?

The retained pipeline is:

1. Cross-session ROI registration between NeutralExposure and offline sessions, using confidence-filtered spatial matching to define candidate unit identities across days [2], [3].
2. Ensemble extraction from the NeutralExposure session using NMF with `k = 8`, followed by definition of core units as the top `15%` of weights within each ensemble [4].
3. Detection of offline high-synchrony events from the offline session using event-first population synchrony criteria, with event windows compared against duration-matched inter-event windows from the same session [5], [6].
4. Computation of a session-level enrichment statistic for each event-ensemble pair: active core-unit fraction during the event minus the average active fraction across ten matched inter-event windows.

Robustness checks include an activity-threshold sweep (`0.0`, `0.5`, `1.0 sigma`) and registration-shuffle controls. The result that survives these checks is modest but consistent: NeutralExposure-defined core units are somewhat more active during detected offline events than expected from matched inter-event baseline, yet much of the absolute signal remains attributable to the general population-burst character of those events.

Operationally, that retained claim sits on top of two nested nulls. First, each event is contrasted against ten duration-matched inter-event windows from the same offline session. Second, the PRI score within each event is itself standardized against a size-matched random-core null (`null_n = 500`), and the full real-registration result is compared against `20` registration-shuffle repeats. The main-text claim is therefore not "events are active", but "real ensemble-core mappings are modestly more enriched than matched-window and shuffled-registration controls."

Pair count	**Threshold sweep**	**Shuffle control**	**Retained claim**
3 NeutralExposure-to-offline pairs	positive enrichment at `0.0`, `0.5`, `1.0 sigma`	real mappings exceed registration-shuffle baseline, but modestly	offline core-unit enrichment is positive but sits above a strong generic burst background

Table S3.1: Compact robustness summary for the DANDI `000718` offline core-unit enrichment analysis.

1.5. DANDI `000336`: Cross-plane access-constraint pipeline

The `000336` analysis [7] targets a different question: do paired imaging planes communicate above null while remaining less coupled than units within the same plane?

The retained full-bundle pipeline is:

1. Load all six NWB files and organize them as three within-session pairs: two primary cross-depth pairs and one supplementary cross-area pair.
2. Separate spontaneous from stimulus-condition windows and use exact timestamp alignment when the two planes share acquisition timing; for the supplementary `ses-1245548523` cross-area session, use coarser `0.5 s` binning because timestamps are interleaved rather than shared.

3. Merge short stimulus presentations into block-level windows before computing pairwise correlations, while retaining a conservative minimum block duration rule.
4. Compare observed cross-plane coupling against within-plane coupling and against within-window circular-shift nulls (`n = 200`).

Under this conservative pipeline, spontaneous cross-plane coupling is above null in all three pairs (`z = 4.04` to `5.16`) and remains above null across all tested stimulus conditions. The stricter `cross < both within` criterion is fully met in the supplementary `ses-1245548523` cross-area pair, but only partially met in the two cross-depth pairs because one within-plane estimate is unusually weak or unusually strong. The retained claim is therefore bounded: the full bundle supports structured above-null inter-plane coupling across all analyzed bundle pairs and tested conditions, while only the supplementary cross-area pair supplies a clean bilateral access-constraint match.

The crucial statistical point is that the `000336` null preserves single-plane temporal structure. Circular shifts are performed separately within each window and only then pooled with window-length weighting, so the `z` scores test whether the two planes align more strongly than expected from their own autocorrelation and rate structure alone, not merely whether the raw traces are non-flat.

Pair	**Geometry**	**Spontaneous z**	**All tested conditions above null?**	**Strict `cross < both within` verdict**
pair_a	cross-depth	4.98	yes	partial
pair_b	cross-depth	4.04	yes	partial
pair_c	cross-area	5.16	yes	positive

Table S3.2: Compact robustness and claim-boundary summary for the DANDI `000336` full-bundle coupling analysis.

1.6. DANDI `001710`: Context-sensitive place-code and remapping baseline

The `001710` analysis [8] targets a third question: do longitudinal hippocampal population codes show the combination of stable spatial tuning, within-day arm separation, and partial cross-day continuity expected from a context-sensitive accessibility account, and does perturbing a candidate postsynaptic structural-write mechanism reduce that continuity selectively across days?

The retained group-bundle pipeline is:

1. Load the broad `23`-subject longitudinal bundle (`7` Cre, `9` Ctrl, `7` SparseKO; `139` NWB files total) and verify readiness with the generic `dandi_io` inventory and probe outputs.
2. Generalize the I/O layer so that both single-channel and multi-channel sessions resolve behavior and ophys containers without hardcoded interface names.
3. Reconstruct trials from the 2P-aligned behavior channels (`trial start`, `trial end`, `trial number`, `block`, `left or right`) cross-referenced against the embedded `trial_cell_data` annotation blob.
4. Extract conservative `df` matrices together with aligned behavior, then compute occupancy-normalized tuning curves for each ROI separately within each session or channel.
5. Compute subject-level observables: within-day left-versus-right arm population-vector structure, cross-day similarity from index-matched ROI tuning curves, subject-level permutation nulls for group comparisons, and genotype-aggregated day-lag profiles. For SparseKO subjects, the canonical group comparison uses `ch0` as a predefined single-channel summary so

that each animal contributes one observation to the genotype contrast, with channel-level robustness summarized separately below.

Under this broad group-bundle pipeline, all `139` files were locally available and entered the QC sweep. Subject-level similarity summaries retained `23` subjects in total: `18` with full `6`-day coverage, `4` with `5` usable days, and `1` with `4` usable days. Split-half reliability remained high across groups, and the retained cross-day means were `0.3374` for Cre, `0.2926` for Ctrl, and `0.2623` for SparseKO. The broad retained claim is therefore stronger than in the original replication bundle: lower cross-day stabilization in SparseKO is now supported at the subject-group level rather than only by a single-animal bridge.

A dedicated follow-on robustness run (`experiments/dandi_001710_08_robustness_and_nulls.py`) added four checks. First, the arm-label audit found `vr_trial_info` content but not a clean per-trial arm truth table, so arm counts remain a QC sanity check rather than a direct label-validation result. Second, the subject-level group null tests placed SparseKO below Cre but not cleanly below Ctrl. Third, the lag profile remained lower for SparseKO across lags `1` to `5`, arguing against the effect being driven by a single interval. Fourth, SparseKO channel comparison showed quantitative sensitivity rather than a qualitative reversal: channel `1` averaged somewhat higher cross-day stability and slightly stronger arm separation than channel `0`, while both channels remained high-reliability. The main-text use of `ch0` should therefore be understood as a predefined bookkeeping choice for one-channel-per-animal comparison, not as evidence that `ch0` is uniquely privileged biologically.

The permutation null behind that claim is fully empirical. The observed group-mean difference is recalculated after `1000` label shuffles that preserve the original genotype group sizes, generating a null distribution of mean differences rather than relying on Gaussian assumptions at the subject level. This matters because the bundle is moderate in size and partially unbalanced (`7` SparseKO, `7` Cre, `9` Ctrl), so the permutation framework is the most transparent way to show that the SparseKO-versus-Cre separation is not a bookkeeping artifact.

Group	**Subjects**	**Usable days / subject**	**ROI range**	**Split-half reliability**	**QC / interpretive note**
Cre	7	5-6	250-1840	0.949-0.986	highest stability mean; clean reference group
Ctrl	9	4-6	639-1694	0.817-0.984	broader heterogeneity but still usable baseline
SparseKO	7	5-6	ch0: 140-504; ch1: 31-199	ch0: 0.884-0.980; ch1: 0.951-0.992	lowest canonical stability mean; channel-sensitive rather than channel-broken

Table S3.3: Compact QC summary for the broadened DANDI `001710` genotype bundle, highlighting group coverage, ROI ranges, and split-half reliability before claim interpretation.

Comparison	SparseKO mean	Other mean	SparseKO n	Other n	z / p	Claim
SparseKO vs Cre	0.2623	0.3374	7	7	`-2.1495 / 0.009`	SparseKO lower than Cre; null-separated
SparseKO vs Ctrl	0.2623	0.2926	7	9	`-1.2856 / 0.099`	directionally lower, but not null-separated

Table S3.4: Subject-level permutation null tests for the broadened DANDI `001710` genotype comparison. SparseKO falls below Cre under the implemented subject-level null and is directionally lower than Ctrl, but the latter contrast is weaker.

Metric	ch0	ch1	ch0 - ch1	Interpretation
Cross-day similarity	0.2623	0.3296	−0.0673	channel `1` is somewhat more stable, but the effect is not reversed
Within-day arm separation	0.2796	0.3119	−0.0323	both channels preserve within-day structure; `ch1` is slightly stronger on average
Split-half reliability	0.9602	0.9767	−0.0165	both channels remain high-reliability

Table S3.5: SparseKO channel comparison for the broadened DANDI `001710` robustness pass. The two channels differ quantitatively but not qualitatively: both support preserved within-session structure, while channel `1` is somewhat more stable across days on average.

Cohort	lag1	lag2	lag3	lag4	lag5
Cre	0.3445	0.3299	0.3307	0.3265	0.3634
Ctrl	0.2992	0.2931	0.2924	0.2819	0.2785
SparseKO	0.2812	0.2766	0.2555	0.2340	0.1973

Table S3.6: Group-level day-lag profile for the broadened DANDI `001710` robustness pass. SparseKO remains lower than the non-KO cohorts across all tested day-lags, arguing against the stability deficit being driven by a single recording interval.

1.7. Repository entry points

The public GitHub repository and archived Zenodo snapshot are structured so that reviewers can inspect the exact scripts and configs used for these observable bridges. Use `reviewer_slow_branch_level_accessibility.ipynb` and `CLAIMS_TO_EXPERIMENTS.md` for the top-level routing; the main S3-specific entry points are:

- `configs/dandi/dataset_000718.yaml`
- `configs/dandi/dataset_000336.yaml`
- `configs/dandi/dataset_001710.yaml`
- `configs/dandi/dataset_001710_replication_bundle_01.yaml`
- `configs/dandi/dataset_001710_group_bundle_01.yaml`
- `python -m dandi_io.cli list --config <yaml>`
- `python -m dandi_io.cli download --config <yaml>`
- `python -m dandi_io.cli probe --config <yaml>`
- `experiments/dandi_000718_14_h1_pri_enrichment.py`
- `experiments/dandi_000336_04_sub656228_replication.py`

- `experiments/dandi_000336_05_ses1245548523.py`
- `experiments/dandi_000336_06_full_bundle.py`
- `experiments/dandi_001710_03_trial_reconstruction.py`
- `experiments/dandi_001710_05_within_day_place_tuning.py`
- `experiments/dandi_001710_06_cross_day_remapping_baseline.py`
- `experiments/dandi_001710_07_replication_bundle_full_pass.py`
- `experiments/dandi_001710_08_robustness_and_nulls.py`

The resulting artifacts are written under `data/dandi/triage/000718`, `data/dandi/triage/000336`, and `data/dandi/triage/001710`, including the `000336` full-bundle coupling outputs, the `001710` replication-bundle and group-bundle outputs, and the dedicated `001710/robustness/` outputs for null tests, QC summaries, day-lag profiles, and channel comparisons. This makes the empirical claims in the manuscript inspectable at the level of scripts, configs, and output files rather than only at the level of prose summary.

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